Subject Code	Subject Name (Lab Oriented Theory course)	Category	L	T	P	C
CS23532	Computer Networks	PC	3	0	4	5
For B.E. Programme CSE, CSECS, CSD and B.Tech. Programme in IT, AIML & AIDS						

Objectives:

- Understand the concepts of computer networks and error detection-correction of data.
- Be exposed to various addressing schemes and routing protocols.
- Learn the Transport Layer, flow control and congestion control algorithms
- Be familiar with real time applications of networking devices and tools.
- To configure different devices and trace the flow of information between nodes in the network using various tools

UNIT I	**FUNDAMENTALS AND DATA LINK LAYER**	**9**
Building a network – Requirements – Layering and protocols – Internet Architecture – Network software – Application Programming Interface (sockets) – Performance – Link layer Services - Framing - Error Detection and Correction – Reliable Transmission		
UNIT II	**MEDIA ACCESS AND INTERNETWORKING**	**9**
Media Access Protocols – ALOHA - CSMA/CA/CD –Ethernet – Wireless LANs - 802.11- Bluetooth - Switching and Forwarding - Bridges and LAN Switches – Basic Internetworking- IP Service Model – IP Fragmentation - Global Addresses – ARP– DHCP – ICMP		
UNIT-III	**ROUTING**	**9**
Routing – Network as Graph - Distance Vector – Link State – Global Internet –Subnetting - Classless Routing (CIDR) – BGP– IPv6 – Multicast routing - DVMRP- PIM – Case Study-Software Defined Networking (SDN)		
UNIT IV	**TRANSPORT LAYER**	**9**
Overview of Transport layer – UDP – TCP - Segment Format - Connection Management – Adaptive Retransmission – TCP Congestion control – Congestion avoidance (DECbit, RED) – QoS – Application requirements.		
UNIT-V	**APPLICATION LAYER**	**9**
E-Mail (SMTP, MIME, POP3, IMAP), HTTP – DNS - FTP - Telnet – Web Services – SNMP - MIB – RMON– SIP		
	Total Contact Hours :	**45**

	List of Experiments
1	Basic Networking Commands in Linux and Windows operating systems
2	Learning and Assignment of IP Address manually to computers
3	Study of different types of Network Cables and Crimping of cable with RJ45 connector
4	Implement packet sniffing using raw sockets in python
5	Develop a customized ping command to test the server connectivity
6	Building anonymous FTP Scanner using ftplib module
7	Develop a simple calculator using XMLRPC
8	Develop a program to create reverse shell using TCP sockets
9	Design a simple topology and configure with one router, two switches and PCs using Cisco Packet Tracer
10	Customize Switch with Network Modules using Cisco Packet Tracer
11	Examine Network Address Translation (NAT) using Cisco Packet Tracer
12	Nmap to discover live hosts using ARP scan, ICMP scan, and TCP/UDP ping scan in TryHackMe Platform.
13	Demonstrate network forensics using PcapXray tool
14	To capture, save, and analyze network traffic on TCP / UDP / IP / HTTP / ARP /DHCP /ICMP /DNS using Wireshark Tool
15	To Analyze the different types of servers using Webalizer tool
	Contact Hours: 60
	Total Contact Hours: 105

Ex: 1:

LINUX COMMANDS AND WINDOWS COMMANDS

	Task	Linux	Windows
1,	Show Current directory	Pwd	cd
2,	List files	ls	dis
3,	Change directory	cd folder name	cd folder name
4,	Create folder	mkdir folder name	mkdir folder name
5,	Delete files	~~rm~~ file.c	del file.c
6,	Copy file	cp ac bc	copy a.c b.c
7,	clear screen	clear	cls
8,	Display file Content	cat file.c	type file.c

Linux Commands and windows Commands.

Aim:

To study and execute various linux commands for file and directory management text processing and basic networking operation.

→ Study of various Commands used in linux Pwd.

Description: Prints current working directory.

output:

$Pwd

/home/user

Description: Create a new folder.

output:

$ mkdir foldername

$ ls

Desktop documents Downloads foldername

Description: changes the current directory.

output:

$cd foldername

Description: Creates an empty files

output:

$ touch example.txt

$ ls

Description: Writes text into a file.

output:

```
$echo "Hi" > example.txt.
```

Description: Displays the content of a file.

output:

```
$cat example.txt
Hi
```

Description: Searches for a specific pattern in a file.

output:

```
$ grep "H" example.txt
Hi
```

Description: Displays the number of lines, words and bytes in a file.

output:

```
$wc example.txt
1 1 3 example.txt.
```

Description: Counts the number of lines in a file.

output:

```
$wc -l example.txt.
1 example.txt.
```

Description: Counts the number of byts in a file

output:

$wc -c example.txt

3 example.txt.

Description: Counts the number of characters in a file

output:

$wc -m example.txt

3 example.txt.

Description: Display the length of the largest line in a file.

output: $wc -L example.txt

2 example.txt

Basic Networking Commands.

Description: Display the systems hostname

output: $hostname

localhost - line.

Description: Display the routing table.

output: $route

Kernal IP routing table.

Description : shows network Communication & routing information

output : $ netstat

Active internet Connections.

Description : Display the IP address and network interface details.

output : $ ip address show

1: Lo

2 : es pas.

Description : Finds the IP address of domain.

output :

$ns lookup www.google.com

name www.google.com

Addres : 142.250.xxxxx,

Description : Display network interface Configuration.

output :

$ ifConfig

en pas 3

inet 10.02.15

Ping.

Description: Display the ARP Counts.

output:

```
$ arp -a
gateway (10.0.2.2)
```

Result:

Thus the linux and windows Commands are studied and executed succesfully.

Ex: 2 23/07/26

Learning and Assignment of IP Address Manually to the Computer

Aim:

To learns IP addresing and manually assigns IP addres to a Computer. To verify the assignment IP addres using Linux networking Commands.

Theory:

An IP addres is a unique numerical addres assigned to every device connected to a network It enables communication between device over a network. An IPV4 address Consists of four octts seperated by dots (eg: 192:168.1.10)

An IP addres can be assigned either manually (static IP) an automatically (Dynamic) using DHCP. A manually assigned IP addres remains unless changed by user. Along with the IP address, Parameters such as subnet mask DNS are also Configured for Proper network Communication.

Command and sample outputs:

1, Display all IP addres

Command:

ip addres show

sample output:

2: ens 160:

inet 172.16.211.128/24

inet6 fe80::20c:29ff::fe75:9927/64

observation:

Display the IPN4 and IPV6 addres of the networks.

2, Display a specific network interface

Command:

ip addres show dev ens160

sample addres

Output: 2: ens160:

inet 172.16.211.128/24

3, Display Routing Table.

Command:

ip route show

Sample output: default via 172.16.211.2 dev ens160

172.16.211.0/24 dev ens160

observation:

Display the routing table and the default gateway.

4, Display Neighbour (ARP) Table:

Command:

ip neigh show

Sample output:

172.16.211.1 REACHABLE
172.16.211.254 STABLE

observation:

Display the neighbouring device and their MAC address.

Result:

The IP address was manually assigned and verified successfully using Linux networking Commands.

Ex: 3:

Study of different type of networks cables and Crimping of Cable with RJ45 Connection.

23/1/26

Aim:

To study different types of network cables and understand and the crimping process using and RJ45 Connection.

Concept:

Network cables are used to connect Computer and other networking devices for data Communication the most commonly used cable in LAN is the unshielded twited Pair (UTP) cable. AW RJ45 Cable of established a network Connection is attached to end of the cable wire are arranged according to the T568A OV T568B wiring standard before Crimping, and the Completed Cable is tested to ensure Proper Connectivity.

Types of Network Cables:

	Cable types		Description
1,	UTP - (Unshielded twisted pair)	-	Commonly used for ethernet LAN's and office networks.
2,	STP (shielded twisted pair)	-	used in area with high electrical electromagnetic interface.
3,	Coaxial Cable	-	Used for cable television broad band intanet and CCTV system.
4,	Optical fiber cable	-	Used for high, speed high based -width and long distance communication.

Crimping Procedure:

1, Cut the UTP cable to the required length
2, Strip about = -3cm of the outer insulation.
3, Untrist and straighten the wires.
4, Arrange the wire according to T568B standard.

5. Trim the wire and evenly.
6. Insert the wire into the RJ45 connector.
7. Crimp the Connector using a crimping tool.
8. Test the cable using a LAN Cable tester.

Observation:

* the wires were arranged according to the T568B standard.
* The RJ45 Connector was crimped succesfully.
* The Cable tester indicates proper continuity.
* The cable was ready for network Communication.

Result:

The different types of network cables were studied, and the UTP cable was succesfully terminated using and RJ45 Connection and tested for proper Connectivity.

24/7/26 Ex:5:

Develop a Customized ping Command to test server Connectivity

Aim:

To Verify network Connectivity and measure the Connection Performance between two devices.

Questions:

1, Why we are using Ping Commands?

→ To isolate Network Problems (Troubleshooting)

→ To Measure Connection Speed (Latency / Lag)

→ To check the stability of the Connections.

→ And to Monitor the servers by Automated testing.

2, Purpose of Ping Commands:

Reachability → checks the target server is Powered on, Connected to network and Accepting requests.

Latency → Measures the Round-Trip time (RTT) and Targets the delay.

Packet Loss → Detect the data packets lost and tells the user.

Why its established?

→ Ping was created to give network engineers a lightweight, instant, universal diagnostic method.

→ The simplicity and efficiency of the Commands execution.

Steps to Reach the target:

1, DNS Resolution:

The computer queries a DNS Server to Convert the ping Command into IP address.

2, Local ARP Request:

→ Address Resolution Protocol is used when the IP is in local Wi-Fi / LAN

3, Transmitting Request:

→ Network adapter sends the echo request onto the internet.

Procedure:

1, Basic Ping Command:

Ping www.google.com

output: Displays the IP address of server, the number of packets sent, received, lost and round trip-time (RTT)

2, Ping localhost Command:

ping localhost.

output: Replies from 127.0.0.1 Confirm that the TCP/IP stalk on local system is working Properly.

3, Ping Using IP Addres Command:

Ping 8.8.8.8

output: Display connectivity and delay time.

4, Contineous Ping test.

Ping Google.com -t.

output: Continuously pings the server.

5, Ping with specific number of request.

Ping google.com -n 5

output: 5 echo requests and display results.

6, Packet size specification with Ping.

Ping www.google.com -l 1000.

output: Sends a Ping request with Packet size of 1000 bytes.

Result:

The Ping Commands were successfully executed in the windows Command Prompt. The network Connectivity, latency and packet transmission status between local Computer and various servers were verified.

Ex:4

Design a Simple Technology And Configure with one Router. Two Switches and Pcs using CISCO Packet Tracer

Aim:

To design and Configure a simple network technology Consisting of one router, two switches and multiple Pcs in Cisco Packet Tracer, and enable Communication b/w Pcs

Definitions:

→ Routers: A networking device that forwards data Packets between computer networks, directing trafic on the network.

→ Switch: A device that connects devices within a network and uses a MAC addresses to forward data to correct destination.

→ Pc (Personal Computer): End devices in network that users interact with to send & receive data.

→ Cables:

i, Straight-through Cable: Used to Connect different device like Pc to switch or switch to Router.

ii, Crossover cable: Used to connect similar devices like switch to switch (rarely needed with modern switches)

iii, Console Cable: Used to Connect Pc to router or switch Console Port for configuration.

Experiment needed:

-> Cisco packet Tracer software
-> 1 Router (eg: 2811)
-> 2 Switches (eg: 2960)
-> 4 PCs
-> Appropriate cables (Copper straight-Through)

Step-by-step Procedure:

Step:1: Open cisco Packet Tracer.

-> Launch Cisco Packet tracer on your Computer.

Step:2: Add devices to workshop.

-> From the bottom device toolbar, drag and drop:

* 1 Router (1941)
* 2 switches (2960)
* 4 PCs (2 Connected to each switches)

Step:3: Connect devices with cables

Use copper straight-Through cable to connect.

* Router Gigabit Ethernet 0/0 to switch 0 Fast ethernet 0/1
* Router Gigabit Ethernet 0/1 to switch 1 Fast ethernet 0/1.

PCs to switch:

-> PC0 to switch 0 fast ethernet 0/2
-> PC1 to switch 0 fast ethernet 0/3
-> PC2 to switch 1 fast ethernet 0/2
-> PC3 to switch 1 fast ethernet 0/3.

Step:4: Configure Router interface:

-> click the Router
-> click tab
-> Enter the IP address and give no shutdown Command.

Step:5: Configure PC IP address

* Click on each PC
* Desktop tab
* IP Configuration and asign. IP address and gate ways.

For eg: PC0. 192.168.1.10 255.255.255.0

Step:6: Save and Test Connectivity:

→ Save the Packet Tracer file
→ Test the connectivity using Ping Command from PC0 to PC_2 or PC_3.
→ Go to CMD type Ping: 192.168.2.1.0.
→ You should see replies if the network is connected succesfully.

Result:

The PCs connected to different switches can Communicate with each other via the router. The succesfull ping replies verify that the network is Properly Configured and operational.

Ex: 10:

Customize switches with Network Modules using Cisco Packed Tracer.

Aim:

To learn how to add a network module to Cisco switches, connect multiple Pcs and a Server. assign Ip addresses and test ~~Con~~ Connectivity in Cisco Packet Tracer.

Description:

This experiment is about building a simple computer network using a switch with extra module ports. You will Connect five Pcs and one server. assign IPs and chack it all devices can Communicate. This helps you understand how modular switches are used to create flexible networks.

Definition:

A network module is an extra components you Can add to a switch. It gives you more ports or add new types of Connections. In packet Tracer, modules such as HWIC-4ESW let you expand the switch capabilities.

Procedure:

1. open Cisco Packet Tracer and set up your workspace.
2. Drag and drop a 2960-24TT switch to centre.
3. Add five Pcs and one Server-PT from the end device Pannel.

4, Power off the switch:
- click the switch, open Physical tab, turn off.

5, Install HWIC-4ESW module to switch
- In Physical tab, drag HWIC-4ESW into an empty module slot.

6, Power on the switch and let it boot.

7, Connect all devices using Copper Straight-Through cables
PCo → Switch fast ethernet 0/1.

8, Assign IP address to each devices.

9, Test the Connectivity (from any PC open CMD and Ping other devices)

output:

A working network setup where every PC and the server can succesfully Ping each other. Proving that both build-in and module Ports on the switch are functioning.

Result:

The experiment demonstrates succesful Customization of switch with a network module and Connectivity between all devices. The network works as expected and all devices Communicate.

Exp: 4

Date:

Implement Packet Sniffing using raw sockets in Python

Aim:

To implement Packet Sniffing using raw sockets in Python and Capture network Packets Passing through network interface.

Objective:

To understand how raw sockets works how network packets can be Captured & analyse using Python Programming.

Theory:

1, Packet Sniffing is the Process of monitoring & Capturing data Packets travelling across a network.

2, A packet sniffer intercepts Packets & analysis their contents.

3, Raw sockets Provide direct access to lower-level network Protocols, allowing applications to Capture Packets.

4, In Python, the socket Modules can be used to Create raw sockets for Packet Capture.

5, This Technique is useful for networking monitor trouble shooting and security Analysis.

Program:

```
import Socket:
    Sniffer = Socket.Socket (Socket AF_INET, Socket,
                                    SOCK_RAW),
    Siniffer.bind((host,0)).
    while True:
        raw_data, addr = Sniffer.recv from (65535)
    Print ("Packet received from", Addr)
    Print (raw_data).
```

output:

Packet received from:
('192.168.15',0)
b'E \x00\x00C...,

Packet received from:
('192.168.1.10',0)
b'E\x00\x00C.

Result:

Thus Packet sniffing was successfully implemented using raw sockets in Python & network Packets were captured and displayed.

Exp: 7:

Date:

Develop a Simple Calculator Using XMLRPC

Aim:

To develop a simple calculator application using XML-RPC Protocol in Python that Performs Basic arithmetic operations like addition, subtraction, multiplication and division through client-server Communication.

Description: XML-RPC (eXtensible Markup Language Remote Procedure call)

It is a Protocol that uses XML to encode request and responses and HTTP as transport mechanism. It enables communication between a client and a server where the client invokes method hosted on server remotely.

-> Server Holds arithmetic Functions

-> client Connects to the server to perform operations like addition, subtraction, Multiplication and division.

Code:

-> Server-Side Code (Server.Py)

```
From xmlrpc.Server import SimpleXMLRPCServer
def add(x,y):
    return x+y.
```

Software requirements:

→ OS: Windows / LINUX
→ Python Version: 3x or above
→ Modules Required: xmlrpc.server, xmlrpc.client (built-in with Python)
→ Text editor (or) IDE: IDLE / VS CODE / Pycharm.

Procedure:

1, Open a Python (or) Terminal
2, Create two separate Python files server.Py & client.Py
3, Run server.Py file first to start server
4, Run client.Py to Perform arithmetic operations
5, Observe the results displayed on client screen and verify server output.

Output: Server side

XML-RPC server is running on Port 8000...

Client-side:

Inputs: 30 and 10
Addition: 40
Subtraction: 20
Multiplication: 300
Division: 30

```
def sub(x,y):
    return x-y
def Multi(x,y):
    return x*y
def divide (x,y)
    if y! =0
        return x /y
    else :
        return "Error: division by zero".
Server = SimpleXMLRPC Server ("local host" 8000, log Req=0)
Server.register_function (add, "add")
Server . register_function (sub, "sub")
Server . register_function (Multi, "Multi")
Server . register_function (divide, "divide")
Print ("XMLRPC Calculator server is running on Port 8000")
Server . Serve_forever ( ) .
```

→ client - Side Code (client.Py)

```
import xmlrpc . client
Proxy = xmlrpc . client . Serverproxy ("http : //localhost :8000/")
Print ("XMLRPC Calculator Test")
num1, num2 = 30, 10 .
Print ("Iput 30, 10")
Print ("Addition(+) : " Proxy. add (num1, num2))
Print ("Subtract (-) : ", Proxy . Sub(num1, num2))
Print ("multiplication(*) : " Proxy . Multi(num1, num2))
Print ("Divide (/) :", Proxy . divide (num1, num2))
```

Result: The XML based calculator successfully implemented The client communicated with server using XML-RPC, Performed all arithmatic operations remotely and displayed accurate results.